从空气中造面包

The Bread Made from Air



同一种化学工艺，为何既能养人也能助战?

READ 纯净阅读

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RECALL 回想巩固

英文原文

In July 1909, BASF envoys gathered round a squat machine on a bench in Fritz Haber's Karlsruhe laboratory. Behind its gleaming, bolted valves, nitrogen and hydrogen met at crushing pressure and furnace heat, and transparent beads of ammonia began to drip slowly. Haber had pried nourishment from the air.

BASF's Carl Bosch scaled the method up, and from 1913 a gigantic plant at Oppau mass-produced ammonia. Hailed for making bread from air, Haber won the coveted 1918 Nobel Prize. Ploughed into stony soils as fertiliser, ammonia wards off famine as no manure heap could, bringing abundance. By 2008 it was feeding nearly half of mankind.

Haber's ambition had a sinister side. Ammonia stocked Germany's arsenal with nitrates for munitions despite a British blockade, and Captain Haber marshalled chemical warfare. On 22 April 1915 at Ypres, troops vented about 168 tonnes of greenish, poisonous chlorine towards the Allied trenches, the first large-scale gas onslaught.

Haber's wife, Clara Immerwahr, Germany's first woman with a doctoral degree in chemistry, voiced objections to his work. On 2 May 1915, after festivities in his honour, she shot herself in the garden with his service pistol. The next day the widower left for the Eastern Front.

Prestige offered no sanctuary. Of Jewish origin, Haber left Germany an outcast after Hitler's regime seized power, dying in Basel in 1934. His institute's pesticide research had produced Zyklon. A later variant, Zyklon B, was used in Nazi death camps, where relatives of his died.

About half the nitrogen in our proteins and DNA has passed through a Haber-Bosch factory, so each of us carries a little of Haber's chemistry.

中文译文

1909年7月，BASF公司的代表围在弗里茨·哈伯位于卡尔斯鲁厄的实验室工作台旁，看着一台矮壮的机器。在闪亮、用螺栓紧固的阀门后，氮与氢在极高压力和熔炉般的高温下反应，透明的氨液滴开始缓缓滴落。哈伯从空气中撬出了养分。

BASF的卡尔·博施将这一方法扩大到工业规模。1913年起，奥保的一座大型工厂开始批量生产氨。哈伯因“从空气中造面包”而广受赞誉，获得了令人向往的1918年诺贝尔奖。氨作为化肥被施入多石的土壤，以粪肥远不能及的能力抵御饥荒，带来丰收。到2008年，它已养活近半数人类。

哈伯的雄心也有阴暗的一面。尽管英国实施封锁，氨仍为德国军火库提供制造弹药所需的硝酸盐；而哈伯上尉则组织化学战。1915年4月22日，在伊普尔，德军向协约国战壕排放了约168吨黄绿色的有毒氯气，发动了首次大规模毒气攻击。

哈伯的妻子克拉拉·伊梅瓦尔是德国首位获得化学博士学位的女性，她曾反对他的工作。1915年5月2日，在为哈伯举行的庆祝活动之后，她在花园中用他的军用手枪自杀。次日，这位刚刚丧妻的人便前往东线。

声望并未给哈伯提供庇护。身为犹太裔，他在希特勒政权掌权后受到排斥，离开德国，1934年死于巴塞尔。他所在研究所的杀虫剂研究曾催生齐克隆；后来的变体齐克隆B被用于纳粹死亡营，他的一些亲属也死在这些营中。

我们体内蛋白质和DNA所含的氮，约有一半曾经过哈伯—博施工厂。因此，每个人的身体里都带着一点哈伯化学的痕迹。

齐克隆B是后来的变体；文中未认定哈伯亲属的具体死因。

核心词汇

先记这22项。连着下方的原文短语一起回想。

valve n.
阀门
its gleaming, bolted valves

ammonia n.
氨
transparent beads of ammonia

nourishment n.
养分；营养
pried nourishment from the air

scale up 动词短语
扩大规模
scaled the method up

mass-produce v.
批量生产
mass-produced ammonia

coveted adj.
人人渴望得到的
the coveted 1918 Nobel Prize

fertiliser n.
化肥；肥料
as fertiliser

ward off 动词短语
抵御；防止
ammonia wards off famine

famine n.
饥荒
ammonia wards off famine

abundance n.
丰饶；充裕
bringing abundance

sinister adj.
险恶的；阴暗的
a sinister side

arsenal n.
军火库
Germany's arsenal

munitions n.
弹药；军需品
nitrates for munitions

blockade n.
封锁
despite a British blockade

marshal v.
统筹；组织调度
marshalled chemical warfare

onslaught n.
猛烈攻击
the first large-scale gas onslaught

objection to 名词短语
对……的反对意见
voiced objections to his work

sanctuary n.
庇护；避难所
Prestige offered no sanctuary

outcast n.
被排斥的人
left Germany an outcast

pesticide n.
杀虫剂；农药
His institute's pesticide research

variant n.
变体
A later variant, Zyklon B

nitrogen n.
氮
nitrogen and hydrogen met

文中词汇与搭配

补充词汇共38项。保留每个表达在本文中的意思。

envoy n.
代表；使者

squat adj.
矮壮的；矮而宽的

bench n.
实验工作台

gleaming adj.
闪亮的

bolted adj.
用螺栓紧固的

crushing adj.
极高的；压倒性的

furnace n.
熔炉；高温炉

transparent adj.
透明的

bead n.
液体小滴

drip v.
滴落

slowly adv.
缓慢地

pry v.
撬出；费力取得

from prep.
从；由

up adv.
表示规模扩大

gigantic adj.
巨大的

hail for 动词短语
因……而受到赞誉

plough into 动词短语
翻耕埋入；施入土壤

stony adj.
多石的

soil n.
土壤

文中词汇与搭配 续

补充词汇共38项。保留每个表达在本文中的意思。

manure n.
粪肥

heap n.
堆；一堆

mankind n.
人类

ambition n.
雄心；抱负

stock v.
为……储备；充实

with prep.
用；以（stock…with搭配）

vent v.
排放；释放气体

greenish adj.
略带绿色的

poisonous adj.
有毒的

trench n.
战壕

doctoral adj.
博士学位的

festivity n.
庆祝活动

pistol n.
手枪

widower n.
鳏夫；丧妻的人

prestige n.
声望；威望

regime n.
政权

seize v.
夺取；攫取

protein n.
蛋白质

pass through 动词短语
经过；通过

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Behind its gleaming, bolted valves, nitrogen and hydrogen met at crushing pressure and furnace heat, and transparent beads of _____ began to drip slowly.

02 Ploughed into stony soils as _____, ammonia wards off famine as no manure heap could, bringing abundance.

03 Haber's ambition had a _____ side.

04 Prestige offered no _____.

文章主线

- 突破 1909年哈伯用空气里的氮，在高温高压下与氢合成氨；博施将其放大，1913年实现工业量产。
- 养育 氨制成化肥，提高粮食产量，到2008年已支撑近半人类；人体蛋白质和DNA中约一半的氮也经此工艺。
- 战争 合成氨也供应弹药原料；哈伯1915年组织首次大规模毒气攻击，妻子反对其战争工作。
- 命运 妻子后来自杀，他次日仍赴前线；虽获1918年诺奖，犹太裔的他后来仍遭纳粹排斥，1934年死于流亡中。
- 余波 他研究所的杀虫剂齐克隆，后来衍生出齐克隆B；纳粹将其用于死亡营，他的一些亲属也死在营中。

参考答案　1 ammonia 2 fertiliser 3 sinister 4 sanctuary